

Chadlia Jerad



Embedded Systems & Cyber-Physical Systems — Researcher, Practitioner and Enthusiast
Associate Professor, ENSI, University of Manouba · Twice Visiting Scholar, UC Berkeley EECS
(Fulbright, 2016–17)

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🌐 chadliajerad.github.io 📺 chadlia-jerad-53490512 🐙 ChadliaJerad
🎓 Google Scholar

Education

- | | | |
|-------------------|--|-----------------|
| Ph.D. | ENIT, University of Tunis El Manar , Electrical Engineering | TN
May 2008 |
| | <ul style="list-style-type: none"> Thesis: Contribution to the description and formal verification of software architectures based on rewriting logic: Application to real-time systems Distinction: Very honorable Advisors: Kamel Barkaoui (CNAM, Paris, FR), Noureddine Ellouze and Amel Grissa-Touzi (ENIT, University of Tunis El Manar, TN) | |
| DEA (M.S.) | ENIT, University of Tunis El Manar , Automatic Control and Signal Processing | TN
July 2003 |
| | <ul style="list-style-type: none"> Thesis: Deductive databases analysis tool based on extended predicate Petri nets Advisors: Amel Grissa-Touzi (ENIT, University of Tunis El Manar, TN) | |
| M.S. Eng. | ENIT, University of Tunis El Manar , Electrical Engineering | TN
July 2002 |
| | <ul style="list-style-type: none"> National Diploma of Engineering (Diplôme d'Ingénieur), Electrical Engineering. A highly selective 5-year rigorous engineering program (accredited at the master's-in-engineering level) Capstone: Automatic code generation of control programs for Siemens PLCs from PNML (Petri Net Markup Language) descriptions | |

Academic Appointments

- | | |
|---|---------------------------|
| ENSI, University of Manouba , Associate Professor | TN
Sep 2009 – Present |
| <ul style="list-style-type: none"> Teaching: Undergraduate and graduate courses in Embedded Systems, Cyber-Physical Systems, Real-Time Operating Systems, and Computer Architecture Advising: M.S. Eng. projects and capstones, M.S. and Ph.D. theses; thesis committee member Research: Embedded Systems, Cyber-Physical Systems, Time-Predictable Execution, and Timing-Aware Deployment of Learned Controllers (Physical AI) Partnerships: Sagemcom Software & Technologies, Actia Engineering, Sofia Technologies (industry); TU Dresden, HS Offenburg (academic) | |
| ISSATS, University of Sousse , Assistant Professor | TN
Sep 2007 – Aug 2009 |
| ENIT, University of Tunis El Manar , Assistant Professor (Contractor) | TN
Sep 2003 – Aug 2007 |

Visiting Research Appointments

- | | |
|--|----------------------------|
| EECS Department, UC Berkeley , Visiting Scholar | USA
Oct 2022 – Aug 2023 |
| <ul style="list-style-type: none"> Host: Prof. Edward A. Lee Role: Contributed to the Lingua Franca project. Formalized and implemented transient federates. | |

- Host: Prof. Edward A. Lee
- Role: Contributed to the Accessors project. Formalized and implemented mutation and deterministic temporal semantics.

Sep 2016 – Aug 2017

Industry Experience

iCompass (AI Tunisian startup, NLP), Consultant in Embedded Engineering

TN

- Designed and developed the Voice Push platform
- Designed and developed a holographic agent

Nov 2021 – Aug 2022

Department of Electronic Security, Mediterranean Services, Engineer

TN

May – Sep 2003

Publications: Peer-Reviewed Journal Articles

- C. Jerad and E. A. Lee, "Decentralized Coordination in LF: Timing Correctness and Transient Federate Support," *IEEE Embedded Systems Letters*, 2026. [Early access] DOI: [10.1109/LES.2026.3729918](https://doi.org/10.1109/LES.2026.3729918)
- B. Oueslati and C. Jerad, "Runtime Monitoring of Observation-Delay Tolerant Reinforcement Learning Policies in Lingua Franca," *IEEE Embedded Systems Letters*, 2026. [Early access] DOI: [10.1109/LES.2026.3726815](https://doi.org/10.1109/LES.2026.3726815)
- S. Lin, E. R. Jellum, M. Theile, T. Tanneberger, B. Sun, C. Jerad, Y. Xu, G. Feng, M. Mæhlum, J. Chen, M. Schoeberl, L. T. X. Phan, J. Castrillon, S. A. Seshia, and E. A. Lee, "Quasi-Static Scheduling for Deterministic Timed Concurrent Models on Multi-Core Hardware," *ACM Transactions on Embedded Computing Systems*, vol. 24, no. 5s, pp. 1–25, 2025. [Special issue, EMSOFT'25] DOI: [10.1145/3762653](https://doi.org/10.1145/3762653)
- H. Bouraoui, C. Jerad, and J. Castrillon, "Combining Early Exit and Selective Prediction for Convolutional Neural Networks," *IEEE Embedded Systems Letters*, 2025. [Special issue on Time-Centric Reactive Software (TCRS, ESWeek 2025)] DOI: [10.1109/LES.2025.3595439](https://doi.org/10.1109/LES.2025.3595439)
- M. Schoeberl, E. R. Jellum, S. Lin, C. Jerad, E. J. Maroun, M. Lohstroh, and E. A. Lee, "Worst-Case Performance for Real-Time Systems," *Research Directions: Cyber-Physical Systems*, vol. 3, pp. e2, 2025. DOI: [10.1017/cbp.2025.1](https://doi.org/10.1017/cbp.2025.1)
- C. Jerad and E. A. Lee, "Toward Dynamism in Distributed Lingua Franca Programs," *IEEE Embedded Systems Letters*, vol. 17, no. 2, pp. 111–114, 2025. [Special issue on Time-Centric Reactive Software (TCRS, ESWeek 2024)] DOI: [10.1109/LES.2024.3465408](https://doi.org/10.1109/LES.2024.3465408)
- S. Lin, Y. A. Manerkar, M. Lohstroh, E. Polgreen, S. Yu, C. Jerad, E. A. Lee, and S. A. Seshia, "Towards Building Verifiable CPS Using Lingua Franca," *ACM Transactions on Embedded Computing Systems*, vol. 22, no. 5s, pp. 1–24, 2023. DOI: [10.1145/3609134](https://doi.org/10.1145/3609134)
- M. Lohstroh, H. Kim, J. C. Eidson, C. Jerad, B. Osyk, and E. A. Lee, "On Enabling Technologies for the Internet of Important Things," *IEEE Access*, vol. 7, pp. 27244–27256, 2019. DOI: [10.1109/ACCESS.2019.2901509](https://doi.org/10.1109/ACCESS.2019.2901509)
- C. Brooks, C. Jerad, H. Kim, E. A. Lee, M. Lohstroh, V. Nouvellet, B. Osyk, and M. Weber, "A Component Architecture for the Internet of Things," *Proceedings of the IEEE*, vol. 106, no. 9, pp. 1527–1542, 2018. DOI: [10.1109/JPROC.2018.2812598](https://doi.org/10.1109/JPROC.2018.2812598)
- H. Bouraoui, C. Jerad, A. Chattopadhyay, and N. B. Hadj-Alouane, "Hardware Architectures for Embedded Speaker Recognition Applications: A Survey," *ACM Transactions on Embedded Computing Systems*, vol. 16, no. 3, pp. 1–28, 2017. DOI: [10.1145/2975161](https://doi.org/10.1145/2975161)
- A. Louati, C. Jerad, and K. Barkaoui, "Formalization and Verification of Hierarchical Use of Interaction Overview Diagrams Using Timing Diagrams," *International Journal of Soft Computing and Software Engineering*, vol. 3, no. 3, pp. 205–211, 2013.
- A. Grissa-Touzi, C. Jerad, and K. Barkaoui, "Nouvelle Approche pour la Definition et la Manipulation de la Negation par les Programmes Stratifies," *Les Annales Maghrebines de l'Ingenieur*, vol. 2, pp. 587–599, 2005.

Publications: Peer-Reviewed Conference Proceedings

- E. R. Jellum, S. Lin, P. Donovan, C. Jerad, E. Wang, M. Lohstroh, E. A. Lee, and M. Schoeberl, "InterPRET: A Time-Predictable Multicore Processor," in *Proceedings of CPS-IoT Week 2023*, pp. 331–336, 2023. DOI: [10.1145/3576914.3587497](https://doi.org/10.1145/3576914.3587497)
- M. A. Messaad, C. Jerad, and A. Sikora, "AI Approaches for IoT Security Analysis," in *Intelligent Systems, Technologies and Applications*, pp. 47–70, 2021. DOI: [10.1007/978-981-16-0730-1_4](https://doi.org/10.1007/978-981-16-0730-1_4)

- C. Jerad and E. A. Lee, "Deterministic Timing for the Industrial Internet of Things," in *2018 IEEE International Conference on Industrial Internet (ICII)*, pp. 13–22, 2018. DOI: [10.1109/ICII.2018.00010](https://doi.org/10.1109/ICII.2018.00010)
- M. B. M. Masmoudi, C. Jerad, and R. Attia, "On-the-Fly Architecture Design and Implementation of a Real-Time Stereovision System," in *Advanced Concepts for Intelligent Vision Systems (ACIVS 2016)*, pp. 700–711, 2016. DOI: [10.1007/978-3-319-48680-2_62](https://doi.org/10.1007/978-3-319-48680-2_62)
- A. Louati, C. Jerad, and K. Barkaoui, "On CPN-Based Verification of Hierarchical Formalization of UML 2 Interaction Overview Diagrams," in *5th International Conference on Modeling, Simulation and Applied Optimization (ICMSAO)*, pp. 1–6, 2013. DOI: [10.1109/ICMSAO.2013.6552608](https://doi.org/10.1109/ICMSAO.2013.6552608)
- A. Louati, K. Barkaoui, and C. Jerad, "Formalization and Verification of Hierarchical Use of Interaction Overview Diagrams," in *International Conference on Software and Systems Engineering and their Applications (ICSSEA)*, 2012.
- C. Jerad, K. Barkaoui, and A. Grissa-Touzi, "On the Use of Real-Time Maude for Architecture Description and Verification: A Case Study," in *Visions of Computer Science: BCS International Academic Conference*, pp. 7–10, 2008. DOI: [10.14236/ewic/VOCS2008.26](https://doi.org/10.14236/ewic/VOCS2008.26)
- C. Jerad, K. Barkaoui, and A. Grissa-Touzi, "Hierarchical Verification in Maude of LfP Software Architectures," in *European Conference on Software Architecture (ECSA)*, pp. 156–170, 2007. DOI: [10.1007/978-3-540-75132-8_12](https://doi.org/10.1007/978-3-540-75132-8_12)
- C. Jerad, K. Barkaoui, and A. Grissa-Touzi, "On the Design and Analysis of Real-Time Systems," in *22nd International Conference on Computers and Their Applications (CATA)*, 2007.
- C. Jerad, K. Barkaoui, and A. Grissa-Touzi, "Verification des architectures des systemes distribues par utilisation du LTL model checker de Maude," in *8eme Colloque Africain sur la Recherche en Informatique (CARI'06)*, pp. 99–106, 2006.
- C. Jerad, A. Grissa-Touzi, and H. Ounelli, "New Approach for Manipulation of Stratified Programs," in *World Enformatika Conference (WEC'05)*, 2005.
- C. Jerad, A. Grissa-Touzi, and H. Ounelli, "STRPRO Tool for Manipulating Stratified Programs Based on SEPN," in *World Enformatika Conference (WEC'05)*, 2005.

Publications: Workshops

- H. Bouraoui, C. Jerad, and J. Castrillon, "Towards Adaptive Multi-Alternative Process Network," in *12th Workshop on Parallel Programming and Run-Time Management Techniques for Many-core Architectures (PARMA-DITAM 2021)*, pp. 1:1–1:15, 2021. DOI: [10.4230/OASICS.PARMA-DITAM.2021.1](https://doi.org/10.4230/OASICS.PARMA-DITAM.2021.1)
- H. Bouraoui, J. Castrillon, and C. Jerad, "Comparing Dataflow and OpenMP Programming for Speaker Recognition Applications," in *PARMA-DITAM 2019*, pp. 1–6, 2019. DOI: [10.1145/3310411.3310417](https://doi.org/10.1145/3310411.3310417)
- A. Louati, K. Barkaoui, and C. Jerad, "Time Properties Verification of UML/MARTE Real-Time Systems," in *IEEE Workshop on Formal Methods Integration (IEEE FMI)*, pp. 663–670, 2014. DOI: [10.1109/IRI.2014.7051930](https://doi.org/10.1109/IRI.2014.7051930)
- A. Louati, K. Barkaoui, and C. Jerad, "Temporal Properties Verification of Real-Time Systems Using UML/MARTE/OCL-RT," in *Workshop on Formal Methods Integration*, pp. 133–147, 2014.
- C. Jerad, K. Barkaoui, and A. Grissa-Touzi, "Vers une verification hierarchique des architectures logicielles," in *7emes Journees Scientifiques des Jeunes Chercheurs en Genie Electrique et Informatique (GEI'07)*, 2007.
- C. Jerad and K. Barkaoui, "On the Use of Rewriting Logic for Verification of Distributed Software Architecture Description Based LfP," in *16th IEEE International Workshop on Rapid System Prototyping (RSP 2005)*, pp. 202–208, 2005. DOI: [10.1109/RSP.2005.34](https://doi.org/10.1109/RSP.2005.34)
- C. Jerad, A. Grissa-Touzi, and K. Barkaoui, "Definition d'un Generateur de Systemes Experts Base sur les Reseaux a Predicats," in *4emes Journees Scientifiques des Jeunes Chercheurs en Genie Electrique et Informatique (GEI'04)*, 2004.

Publications: Technical Reports & Preprints

- J. Castrillón-Mazo, C. Jerad, E. A. Lee, C. Pagetti, and S. J. Lin, "Tradeoffs in Reactive Systems Design (Dagstuhl Seminar 25091)," *Dagstuhl Reports*, vol. 15, no. 2, pp. 126–157, 2025. DOI: [10.4230/DagRep.15.2.126](https://doi.org/10.4230/DagRep.15.2.126)
- S. Lin, E. R. Jellum, M. Theile, T. Tanneberger, B. Sun, C. Jerad, R. Xu, G. Feng, C. Menard, M. Lohstroh, J. Castrillon, S. A. Seshia, and E. A. Lee, "PretVM: Predictable, Efficient Virtual Machine for Real-Time Concurrency," *arXiv preprint*, arXiv:2406.06253, 2024. DOI: [10.48550/arXiv.2406.06253](https://doi.org/10.48550/arXiv.2406.06253)
- H. Bouraoui, C. Jerad, O. Romdhani, and J. Castrillon, "MAPN: Modeling, Analysis, and Exploration of Algorithmic and Parallelism Adaptivity," *arXiv preprint*, arXiv:2207.07591, 2022. DOI: [10.48550/arXiv.2207.07591](https://doi.org/10.48550/arXiv.2207.07591)

- [C. Jerad](#) and E. A. Lee, "A JavaScript Extension Providing Deterministic Temporal Semantics for the Internet of Things," Tech. Rep., UCB/EECS-2017-136, EECS Department, University of California, Berkeley, 2017. [Link](#)

Teaching

Levels

M.S. Eng. (Diplôme d'Ingénieur) = 2 preparatory years + 3 years of engineering school, accredited at the master's-in-engineering level (Years 3–5); M.S. = 2 years after a 3-year Licence (Year 4 = M.S. 1st year, Year 5 = M.S. 2nd year); Licence = Years 1–3. "Year" counts years after high school.

Cyber-Physical Systems

ENSI, University of Manouba
Current

- M.S. Eng. & M.S., Year 5; Language: EN
- Based on [EECS 149/249A](#) at UC Berkeley
- Hands-on: [embedded labs](#) on the Pololu 3pi+ robot (M.S. Eng.), robot simulator [IR-Sim](#) (M.S.)

Real-Time Operating Systems

ENSI, University of Manouba
Current

- M.S. Eng., Year 5; Language: EN, FR
- Textbook: Liu, J. W. S. (2000). Real-Time Systems. Pearson Education/Prentice Hall.
- Served as class coordinator among involved instructors (2012-2016)

Introduction to Embedded Systems

ENSI, University of Manouba
Current

- M.S., Year 4; Language: EN
- Forked from [EECS 149/249A](#) at UC Berkeley

Sensors and Acquisition (Guest Lecturer)

MSE, University of Manouba
2025 - 2026

- M.S. Eng., Year 3; Language: EN

Embedded Systems (Guest Lecturer)

EPT, University of Carthage
2017 & 2019

- M.S. Eng., Year 4; Language: EN

Computer Architecture (Guest Lecturer)

EPT, University of Carthage
2019 – 2020

- M.S. Eng., Year 3; Language: EN

Sensors, Networks and Internet of Things (Guest Lecturer)

ENIT, University of Tunis El Manar
2017 - 2019

- M.S., Year 5; Language: EN

Advanced Architectures and Parallel Programming

ENSI, University of Manouba
2021 – 2024

- M.S. Eng., Year 5; Language: EN, FR
- Based on [CS 267](#) at UC Berkeley

Formal Methods

ENSI, University of Manouba
2021 – 2022

- M.S., Year 4

Programming Connected Objects

ENSI, University of Manouba
2021 – 2023

- M.S. Eng., Year 5; Language: EN

Introduction to Embedded Systems

ENSI, University of Manouba
2012 - 2019

- M.S. Eng. & M.S., Year 4; Language: FR
- Elaborated in collaboration with ST Microelectronics, TN
- Served as class coordinator among involved instructors

Microprocessor Design

ENSI, University of Manouba
2009 - 2019

- M.S. Eng., Year 4; Language: FR

- Based on [CS 152](#) at UC Berkeley
- Served as class coordinator among involved instructors (2011-2019)

Computer Architecture

- M.S. Eng., Year 3; Language: FR

ENSI, University of Manouba
2009 - 2012

Computer Architecture

- Licence, Year 2; Language: FR

ISSATS, University of Sousse,
TN
2007 – 2009

Digital Circuits

- Licence, Year 2; Language: FR

ISSATS, University of Sousse,
TN
2007 – 2009

Computer Architecture

- M.S. Eng., Year 3; Language: FR

ENIT, University of Tunis El
Manar, TN
2003 – 2007

Algorithms and C/C++ Programming

- M.S. Eng., Year 3; Language: FR

ENIT, University of Tunis El
Manar, TN
2003 – 2007

Talks, Tutorials & Presentations

Tutorial — [Lingua Franca: An Open-Source Coordination Language for Deterministic Integration of Cyber-Physical Systems](#)

May 2026

Presented at CPS-IoT Week 2026, Saint Malo, FR

- H. Kim, C. Jerad, E. A. Lee

Invited Talk — Cyber Physical Systems & Internet of Things; Example Applications and Innovation Frameworks

Jul 2021

Chinese & Tunisian Innovation Challenge, 2nd edition of One Belt One Road: Global Digital Online Study 2021

Invited Talk — Cyber Physical Systems and Internet of Things

Jan 2021

Chinese & Tunisian Innovation Challenge, 1st edition of One Belt One Road 2021

Invited Talk — A browser for Things: using the Accessors software architecture

Jun 2018

ivESK-Kolleg, HS-Offenburg, DE

Invited Talk — A browser for Things: using the Accessors software architecture

Apr 2018

Chair for Compiler Construction, cfaed, TU Dresden, DE

Poster — [Augmented Reality with Accessors](#)

Oct 2017

TerraSwarm 2017 Annual Meeting, Berkeley, CA

- M. Weber, E. A. Lee, C. Brooks, C. Jerad, H. Kim, M. Lohstroh, V. Nouvellet, B. Osyk

Presentation — Running Accessors On Mobile Devices Using Cordova

May 2017

The Ptolemy Group Lunch, Berkeley, CA

- C. Jerad, H. Kim, M. Lohstroh, V. Nouvellet, M. Weber

Presentation — Mutable Accessors & Deterministic Temporal Semantics

Apr 2017

The Ptolemy Group Lunch, Berkeley, CA

Presentation — Tutorial on Maude

Nov 2016

The Ptolemy Group Lunch, Berkeley, CA

Presentation — Bringing Swarmlets into River Styx

The Ptolemy Group Lunch, Berkeley, CA

Sep 2016

Invited Talk — Model checking architectural descriptions: software and hardware

ZIH Colloquium, TU-Dresden, DE

Apr 2015

Software & Artifacts

Lingua Franca: Contributor — a polyglot coordination language for deterministic concurrent and distributed systems.

Accessors / CapeCode: Contributor — a component framework for IoT applications (swarmlets), built on [Ptolemy II](#).

Graduate Advising

Yassine Hattay — Ph.D.

ENSI, University of Manouba

- Topic: Characterization, Modeling and Deadline-Aware Optimization of CPU-Only Vision-Language Model Inference on Edge and Embedded Platforms
- Co-advised with Majdi Jribi (ENSI, University of Manouba)

Ongoing

Hasna Bouraoui — Ph.D.

TU Dresden

- Topic: Adaptivity in deploying applications onto heterogeneous embedded platforms
- Co-advised with Jeronimo Castrillon (TU Dresden)

Ongoing

Balkis Oueslati — M.S.

ENSI, University of Manouba

- Topic: Imitation-Based Reinforcement Learning for Delay-Aware Locomotion: Reward Shaping and Runtime Monitoring for Embedded Deployment

Defended Sep 2026

Ayoub Bnina — M.S.

ENSI, University of Manouba

- Topic: AeroTrack: Active Perception and Closed-Loop Control for UAV Swarm Tracking: Adaptive Early-Exit Inference and Distribution-Aware Association in Photorealistic Simulation

Defended Sep 2026

Yessine Hattay — M.S.

ENSI, University of Manouba

- Topic: CPU-Only LLM Inference: Compute, Bandwidth, and Capacity Limits: A Performance Model for llama.cpp on Consumer and Embedded Hardware

Defended Sep 2026

Omaima Younsi — M.S.

ENSI, University of Manouba

- Topic: Contextual Data Fusion in Collaborative Body Sensor Networks: Application to e-Emergency

Defended Dec 2022

Mohamed Abou Messaad — Ph.D.

ENSI, University of Manouba

- Topic: Towards Low-Cost, Low-Power Hardware Accelerators for Machine Learning-Based Cybersecurity
- Co-advised with Narjes Bellamine (ENSI, University of Manouba) and Axel Sikora (HS Offenburg)

Discontinued 2022 (joined industry)

Abir Hamdi — M.S.

ENSI, University of Manouba

- Topic: Towards Rapid Prototyping and Simulation of mAPN Models
- Co-advised with Jeronimo Castrillon (TU Dresden) and Hasna Bouraoui (TU Dresden)

Defended Dec 2021

Omar Romdhani — M.S.

ENSI, University of Manouba

- Topic: MAPN: Assessment Against State-of-the-Art Dynamic Dataflow MoC
- Co-advised with Jeronimo Castrillon (TU Dresden) and Hasna Bouraoui (TU Dresden)

Defended Dec 2020

Ala Eddine Lajili — M.S.

ENSI, University of Manouba

- Topic: Exploring Mesh-of-Tree Architecture and Developing a Dedicated Router

Defended Jul 2020

Mohamed Taher Aouayeb — M.S.

ENSI, University of Manouba

- Topic: Study of the Impact of Scheduling Frequency on the Time Predictability of FlexPRET Defended Sep 2019
 - Co-advised with Jeronimo Castrillon (TU Dresden)
- Hamza Benaissa — M.S.** ENSI, University of Manouba
 Defended Sep 2019
- Topic: Narrowband IoT Extension for the NS-3 Simulator
 - Co-advised with Axel Sikora (HS Offenburg)
- Mohamed Abou Messaad — M.S.** ENSI, University of Manouba
 Defended Oct 2019
- Topic: Architecture for Embedded Image and Stereo Vision Processing, Based on Chisel
- Moez Othmani — M.S.** ENSI, University of Manouba
 Defended Oct 2019
- Topic: Architecture for Embedded Deep Learning Processing, Based on Chisel
- Mohamed Masmoudi — Ph.D.** EPT, University of Carthage
 Discontinued 2017 (re-
 Stereovision Systems mained in industry)
- Topic: Design Methodologies for Reconfigurable Real-Time Vision Architectures: Focus on Stereovision Systems
 - Co-advised with Rabah Attia (EPT, University of Carthage)
- Nesrine Abid — Ph.D.** ENIT, University of Tunis El
 Manar
 Discontinued 2015 (re-
 mained in industry)
- Topic: Fault Diagnosis Using Hybrid Automata
 - Co-advised with Nejib Ben Hadj Aoulane (ENIT, University of Tunis El Manar)
- Aymen Louati — Ph.D.** ENIT, University of Tunis El
 Manar & CNAM Paris
 Defended May 2015
- Topic: Contribution à la Formalisation et à la Vérification des Diagrammes Dynamiques UML 2 à base des Réseaux de Petri
 - Co-advised with Kamel Barkaoui (CNAM Paris) and Nouredine Ellouze (ENIT, University of Tunis El Manar)
- Beyond the theses listed: about 2.5 M.S. Eng. Year-4 projects and 3 M.S. Eng. Year-5 capstones supervised every year, including award-winning Year-4 projects: best project award in 2017–18 and 2025–26, second best in 2024–25.** ENSI, University of Manouba
 2011 - current

Administrative Responsibilities & Academic Leadership

- ENSI, University of Manouba**, Coordinator, Embedded Software and Systems (SLE) Special-ization TN
 Sep 2014 – Aug 2020
- Directed the academic strategy and updated the 5th-year curriculum
 - Built long-term partnerships with industry and foreign universities, and promoted the specialization to students and companies
 - Organized workshops, seminars, hackathons and info sessions to prepare students for internships and the job market
- ENSI, University of Manouba**, Course Coordinator TN
 Sep 2012 – Aug 2019
- Coordinated multi-instructor teaching teams for three core courses
- ENSI, University of Manouba**, Student Club Advisor & Mentor TN
 2010 – Present
- Faculty advisor for the Robotics Club (Club Robotique ENSI, Association Robotique ENSI), and Open Source Software ENSI Club
 - Mentor students on the technical, logistical and strategic aspects of their projects

Distinctions

- 3D Rationality Fellowship** May 2023 – Aug 2023
- EECS Department, UC Berkeley
 - Supporting the ongoing collaboration between 3D Rationality and UC Berkeley.

Fulbright Visiting Scholar Fellowship

Sep 2016 – Aug 2017

- UC Berkeley, iCyPhy Research Center

Nominated to '100 Future Leaders' (2018, 1st edition)

Mar 2018

- ITES – Tunisian Institute of Strategic Studies

Portrait of the Month

Aug 2017

- DAAD (German Academic Exchange Service) Tunis

Initiatives

Flight Workshop: A Journey through Aeronautics

May 2025

Co-organizer and moderator

- Venue: Technopole de la Manouba, Manouba, TN
- Featuring Dr. Scott Eberhardt

The Cyber and the Physical Symbiosis (CyPhySym)

Mar – Dec 2021

Lead proposer and principal author; Tunisian-side PI · Funded by DAAD (Transformation Program Line 2) · Over €37,000 awarded (partially executed online due to COVID-19)

- Formal applicant: TU Dresden, DE
- Partners: ENSI, University of Manouba, TN
- Organized the thematic session [The Cyber and the Physical Symbiosis](#) with TU Dresden (program, speakers, logistics)
- Funded the research stay of M.S. student A. Hamdi at TU Dresden (co-advised with Prof. J. Castrillon)
- Co-advised the Ph.D. of H. Bouraoui, TU Dresden (with Prof. J. Castrillon)
- Funded reciprocal faculty visits (see Research Stays)

IoT Innovation Challenge (IoTIC)

Mar – Dec 2021

Lead proposer and principal author; Tunisian-side PI · Funded by DAAD (Transformation Program Line 2) · Over €39,000 awarded (partially executed due to COVID-19)

- Formal applicant: HS-Offenburg, DE
- Partners: ENSI, University of Manouba, TN
- Coordinated and mentored Tunisian teams in the Bosch Sensortec IoT Innovation Challenge
- Funded the finalist teams' trip to Stuttgart for the final pitch and award ceremony
- Funded the research stay of Ph.D. student M. Abou Messaad at HS Offenburg (co-advised with Prof. A. Sikora)

Engineering Cyber Physical Systems and Internet of Things (EngCPS&IoT)

Mar – Dec 2020

Lead proposer and principal author; Tunisian-side PI · Funded by DAAD (Transformation Program Line 2) · Over €38,000 awarded (partially executed online due to COVID-19)

- Formal applicant: TU Dresden, DE
- Partners: ENSI, University of Manouba, TN
- Organized the seminar series [Design and Programming Cyber-Physical Systems and IoT Applications](#) (moved online due to COVID-19)
- Funded the research stay of M.S. student O. Romdhani at TU Dresden (co-advised with Prof. J. Castrillon)
- Co-advised the Ph.D. of H. Bouraoui, TU Dresden (with Prof. J. Castrillon)
- Funded reciprocal faculty visits (see Research Stays)

5 Years and More Generations of Fruitful Collaboration in IoT (5G+IoT)

Mar – Dec 2020

Lead proposer and principal author; Tunisian-side PI · Funded by DAAD (Transformation Program Line 2) · Over €39,000 awarded (partially executed online due to COVID-19)

- Formal applicant: HS-Offenburg, DE
- Partners: ENSI, University of Manouba, TN

- Organized the 3rd IoT Hackathon with HS Offenburg
- Funded the research stay of Ph.D. student M. Abou Messaad at HS Offenburg (co-advised with Prof. A. Sikora)

Cooperation for Sustainable Development in Drylands

Mar – Dec 2020

Tunisian-side PI · Funded by DAAD (Transformation Program Line 2)

- Formal applicant: University of Hamburg, DE
- Partners: ENSI, University of Manouba, TN & HS-Offenburg, DE
- Contributed to the 3rd IoT Hackathon (5G+IoT project): the Drylands partners defined the hackathon topic and delivered online talks on it

Embedded parallel programming in practice, teaching and research

Mar – Dec 2019

Lead proposer and principal author; Tunisian-side PI · Funded by DAAD (Transformation Program Line 2) · Over €16,000 awarded

- Formal applicant: TU Dresden, DE
- Partners: ENSI, University of Manouba, TN
- Organized the round table ‘Research landscape and opportunities for technology transfer’ (ENSI, Dec 3, 2019)
- Hosted Prof. J. Castrillon and his Ph.D. students H. Bouraoui and A. Goens (TU Dresden) at ENSI to teach a 15-hour compilers course (M.S. Eng., Year 5)
- Co-advised 1 M.S. Eng. capstone project and the M.S. thesis of M. T. Aouayeb
- Co-advised the Ph.D. of H. Bouraoui, TU Dresden (with Prof. J. Castrillon)
- Funded reciprocal faculty visits (see Research Stays)

Continued Collaboration for IoT (ContIoT)

Mar – Dec 2019

Lead proposer and principal author; Tunisian-side PI · Funded by DAAD (Transformation Program Line 2) · Over €39,000 awarded

- Formal applicant: HS-Offenburg, DE
- Organized the 2nd IoT Hackathon with HS Offenburg
- Organized a study trip to Germany for 10 Tunisian students and 2 faculty members
- Co-advised 1 M.S. Eng. capstone project and the M.S. thesis of H. Benaissa
- Funded reciprocal faculty visits (see Research Stays)

Realizing IoT: in Business and Engineering (RealIoT)

Mar – Dec 2018

Lead proposer and principal author; Tunisian-side PI · Funded by DAAD (Transformation Program Line 2) · Over €27,000 awarded

- Formal applicant: HS-Offenburg, DE
- Partners: ENSI, University of Manouba, TN
- Organized the 1st IoT Hackathon with HS Offenburg
- Hosted Prof. A. Sikora (HS Offenburg) at ENSI to teach a 30-hour IoT course (M.S. Eng., Year 5)
- Co-advised 3 M.S. Eng. capstone projects
- Funded reciprocal faculty visits (see Research Stays)

Get-together towards a sustainable collaboration in IoT

Mar – Dec 2016

Lead proposer and principal author; Tunisian-side PI · Funded by DAAD (Transformation Program Line 2) · Over €28,000 awarded

- Formal applicant: HS-Offenburg, DE
- Partners: TU Dresden (ZIH and cfaed), DE & ENSI, University of Manouba, TN & ENIT, University of Tunis El Manar, TN
- Organized the [IoT School & Workshop 2016](#) across the three partner institutions
- Co-advised 1 M.S. Eng. capstone project
- Funded reciprocal faculty visits

Design and Applications of Multi-Processors System on Chip (Winter School)

Nov 2014

Lead proposer and principal author; Tunisian-side PI · Funded by DAAD (Transformation Program Line 2) · Over €16,000 awarded

- Venue: Cité des Sciences, Tunis, TN
- Formal applicant: cfaed, TU Dresden, DE
- Partners: ENIT, University of Tunis El Manar, TN

Program Committee

- TPC member, Workshop on Reactive Cyber-Physical Systems (ReCPS), co-located with DATE: [2026](#)
- TPC member, IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS): [2026](#)
- TPC member, Design, Automation and Test in Europe Conference (DATE): [2025](#), [2026](#)
- TPC member, Workshop on Time-Centric Reactive Software (TCRS): [2023](#), [2024](#), [2025](#)
- TPC member, IEEE Real-Time Systems Symposium (RTSS): [2021](#), [2022](#), [2023](#)
- Reviewer, [IEEE Access](#), 2023, 2024, 2025

Chairing and Opponent Duties

- | | |
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| PhD Dissertation Committee (Opponent) of Henrik Austad | Sep 2025 |
| <ul style="list-style-type: none"> • Dissertation: ‘Deterministic Networks for Low-Latency Cyber-Physical Systems’ • NTNU, Trondheim, NO | |
| Session Chair: Timing Challenges in Cyber-Physical Systems | Oct 2024 |
| <ul style="list-style-type: none"> • 2nd Workshop on Time-Centric Reactive Software (TCRS 2024) at ESWEEK 2024 | |
| M.S. Thesis Committee (Reviewer) of Bayrem Zarai | Dec 2022 |
| <ul style="list-style-type: none"> • Thesis: ‘Situational Awareness ML-Assisted Unmanned Aerial System Traffic Management’ • ENSI, University of Manouba, TN | |

Editorial and Other Activities

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|--|----------|
| Guest editor of "African Perspectives on Real-Time and Cyber-Physical Systems" | Sep 2026 |
| <ul style="list-style-type: none"> • A special issue of Springer journal Real-Time Systems. | |
| Co-organizing Dagstuhl Seminar 25091: Tradeoffs in Reactive Systems Design | Feb 2025 |
| <ul style="list-style-type: none"> • Schloss Dagstuhl, Wadern, DE • With J. Castrillón-Mazo (TU Dresden, DE), E. A. Lee (UC Berkeley, US), C. Pagetti (ONERA – Toulouse, FR) | |
| Organizing Committee of the 3rd Workshop on Time-Centric Reactive Software (TCRS 2025) | Jan 2025 |
| <ul style="list-style-type: none"> • Co-located with ESWEEK 2025 • Taipei, TW | |
| Dagstuhl Seminar 19101 Participant | Mar 2019 |
| <ul style="list-style-type: none"> • Analysis, Design, and Control of Predictable Interconnected Systems, DE. | |
| Member of the DAAD Scholarship Selection Committee | Jan 2018 |
| <ul style="list-style-type: none"> • German Academic Exchange Service (DAAD) | |

Research Stays

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| cfaed, TU Dresden, DE, Research stay | Aug & Nov 2021 |
| <ul style="list-style-type: none"> • Project: CyPhySym • Funded by: German Academic Exchange Service (DAAD) • Key Achievements: Co-advised 1 M.S. thesis (A. Hamdi) and 1 Ph.D. thesis (H. Bouraoui); co-authored 1 workshop paper (PARMA-DITAM 2021) | |

ivESK, HS Offenburg, DE , Research stay • Project: IoTIC • Funded by: German Academic Exchange Service (DAAD) • Key Achievements: Co-authored 1 conference paper (ISTA 2021), groundwork for the co-advised Ph.D. of M. Abou Messaad.	Sep 2021
cfaed, TU Dresden, DE , Research stay • Project: EngCPS&IoT • Funded by: German Academic Exchange Service (DAAD) • Key Achievements: Co-advised 1 M.S. thesis (O. Romdhani) and 1 Ph.D. thesis (H. Bouraoui); co-authored 1 workshop paper (PARMA-DITAM 2021).	Aug 2020
cfaed, TU Dresden, DE , Research stay • Project: Embedded Parallel Programming • Funded by: German Academic Exchange Service (DAAD) • Key Achievements: Co-advised 1 M.S. thesis (M. T. Aouayeb) and 1 Ph.D. thesis (H. Bouraoui); co-authored 1 workshop paper (PARMA-DITAM 2019).	May & Jul 2019
ivESK, HS Offenburg, DE , Research stay • Project: ContIoT • Funded by: German Academic Exchange Service (DAAD) • Key Achievements: Co-advised 1 M.S. thesis (H. Benaissa).	Jul 2019
ivESK, HS Offenburg, DE , Research stay • Project: RealIoT • Funded by: German Academic Exchange Service (DAAD)	May 2018
CNAM, Paris, FR , Research stay • Funded by: Association MedLink • Key Achievements: Co-advised the Ph.D. of A. Louati (defended 2015); co-authored 5 papers on Petri-net-based verification of UML diagrams (2012–2014).	Jun–Jul 2013
CNAM, Paris, FR , Research stay • Funded by: Ministry of Higher Education, TN • Key Achievements: Doctoral research under the co-advice of Prof. K. Barkaoui.	2004–2006

Technical Growth

ITEC Program: Training Educators on AI Awareness and Technology Integration • NITTTR, Chennai, India (Indian Technical and Economic Cooperation) • 2-week program, 16–29 July 2025	Jul 2025
Introduction to Augmented Reality and ARCore • Google AR & VR (Coursera) • Completed course (w/o certificate), Grade 100%	May 2021
Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning • deeplearning.ai (Coursera) • Certificate, Grade 99.28%	May 2020
LoRa Ecosystem - Networking and Workshop • SUP'COM, TN • 1-day training	Jan 2019

Personal Growth

Interviews: a method to collect data during the research process • Christian Achraimer (DAAD funded) • 1-day training	Oct 2019
Effective Cooperation in Teams • Sonja Drewes (DAAD funded) • 1-day training	Apr 2019
Self-management • Heba Salama (DAAD funded) • 1-day training	Oct 2018
Lean Startup and customer development • Bob Dorf (Open Startup Tunisia) • 2-day training for mentors	Sep 2018
Pre-Emptying & Managing Conflicts • Heba Selim (DAAD funded) • 1-day training	Nov 2017

Volunteering

Student Mentor • Mentoring students at the Hackathon within the International Winter School "IoT meets AI", organized by SUP'COM's IEEE Student Branch	Mar 2020
Student Mentor • Mentoring students: Open Startup Tunisia competition	Oct–Nov 2018
Volunteer • Cragmont Elementary School, Berkeley, CA	Jan 2017
Youth Leader and Coordinator • Astronomy club of the Tunisian Association of Youths and Science (AJST) • Organized astronomy training sessions and public events	1997 – 2005
Founder and Treasurer • "Andromeda Astronomy Club" at ENIT, University of Tunis El Manar, TN	1999 – 2001
Founder and General-secretary • "Astronomy Club" at IPEIT, TN	1997 – 1999